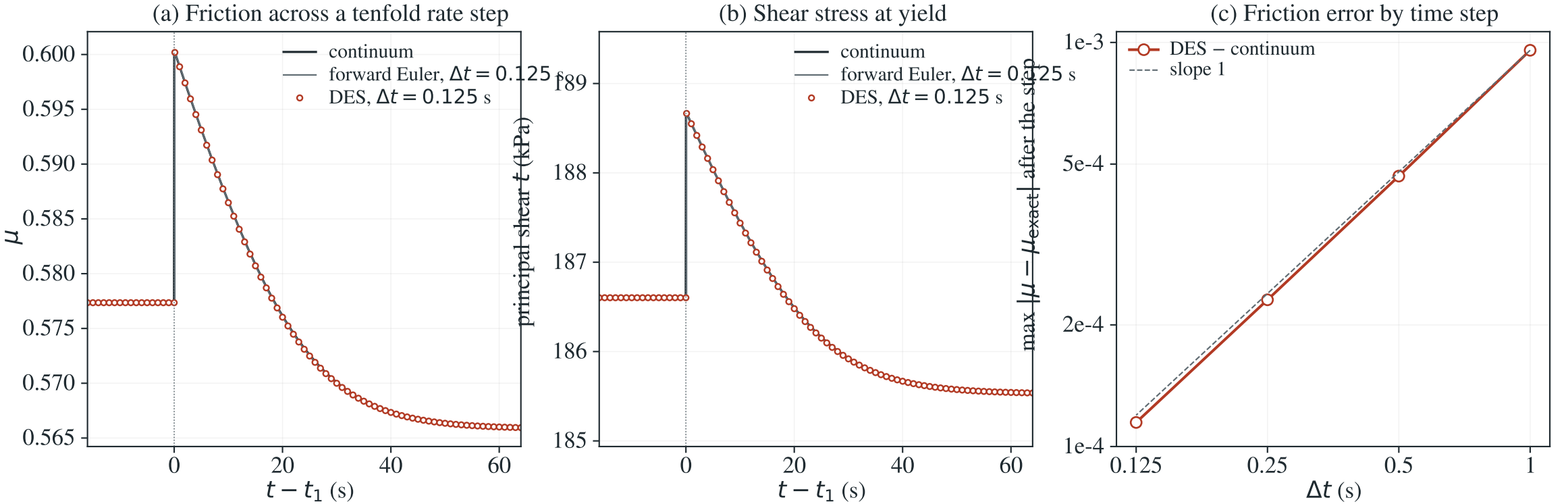

(a) Friction across a tenfold rate step
continuum
forward Euler, $\Delta t = 0.125$ s
DES, $\Delta t = 0.125$ s
μ
0.600
0.595
0.590
0.585
0.580
0.575
0.570
0.565
0
20
40
60
$t - t_1$ (s)
(b) Shear stress at yield
continuum
forward Euler, $\Delta t = 0.125$ s
DES, $\Delta t = 0.125$ s
principal shear t (kPa)
189
188
187
186
185
0
20
40
60
$t - t_1$ (s)
(c) Friction error by time step
DES − continuum
slope 1
max $|\mu - \mu_{\text{exact}}|$ after the step
1e-3
5e-4
2e-4
1e-4
0.125
0.25
0.5
1
Δt (s)